FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2023

TEST DATE: 03-12-2022

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

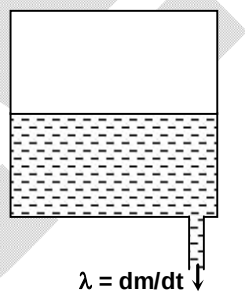
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. A metal vessel of negligible heat capacity contains 0.5 kg of water (of specific heat capacity 4200 $Jkg^{-1}K^{-1}$). It is heated from 15°C to 85°C by 750 W immersion heater in 4 minutes. The average loss of heat in watts to the surroundings from the vessel during this time is nearly:
(A) 70 (B) 123
(C) 138 (D) 212

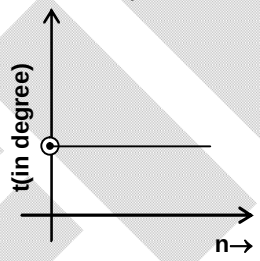
2. A well insulated box contains water (specific heat = C and very large conductivity) of mass m_0 and temperature T_0 at time t = 0. If heat is being added to it uniformly at a constant rate $\frac{dQ}{dt} = R$ and water is leaking from it at a constant rate $\frac{dm}{dt} = \lambda$, then the temperature of the water at time t $\left(t < \frac{m_0}{\lambda}\right)$ will be:

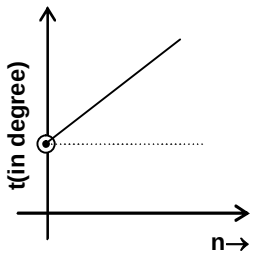


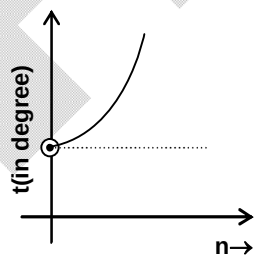
(A) $T = \frac{R}{2\lambda C}\ell n\left(\frac{m_0}{m_0 - \lambda t}\right) + T_0$ (B) $T = \frac{R}{\lambda C}\ell n\left(\frac{m_0}{m_0 - \lambda t}\right) + T_0$

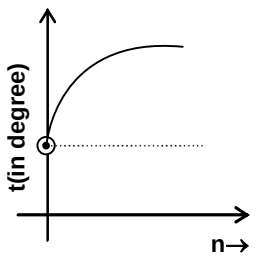
(C) $T = \frac{R}{\lambda C}e^{(m_0 - \lambda t)}$ (D) $T = \frac{R}{2\lambda C}e^{(m_0 - \lambda t)}$

3. A slab X of thickness 't' thermal conductivity 'k' and area of cross-section 'A' is placed in thermal contact with another slab Y which is $2n^2$ times thicker, 4n times conductive and having n times larger cross-section area. If the outside face of X is maintained at 100°C, the outside face of Y at 0°C, then the temperature of the junction is represented by the graph (n > 0) :

(A)


(B)


(C)


(D)


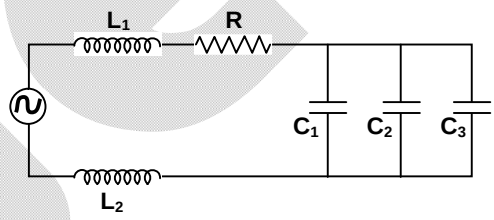
4. Two identical metallic sheets each behaves like a black body are kept parallel to each other with small separation between them in vacuum. Thermal energy is generated at a constant rate P in one of the sheets by passing electricity through it. In steady state, temperature of the other sheet is found to be constant at 300K. Find the value of P in kilo-watt. (given area of plates = $A = \frac{10^3}{17 \times 27} m^2$ and $\sigma = \frac{17}{3} \times 10^{-8}$ W/m^{2}K^4)

(A) 3 (B) 4
(C) 5 (D) 6

5. A copper sphere is suspended in an evacuated chamber maintained at 300K. The sphere is maintained at a constant temperature of 500K by heating it electrically. A total of 300W of electric power is needed to do it. When half of the surface of the copper sphere is completely blackened 600W is needed to maintain the same temperature of the sphere. Calculate the emissivity of copper.

(A) $e = \frac{1}{3}$ (B) $e = \frac{2}{3}$
(C) $e = \frac{1}{2}$ (D) $e = \frac{1}{6}$

6. A generator with an adjustable frequency of oscillation is connected to resistance, R = 100 Ω, inductances, L_1 = 1.7 mH and L_2 = 2.3 mH and capacitances, C_1 = 4 μF, C_2 = 2.5 μF and C_3 = 3.5 μF. The resonant angular frequency of the circuit is

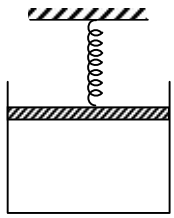


(A) 0.5 rad/s
(B) 0.5×10^4 rad/s
(C) 2 rad/s
(D) 2×10^{-4} rad/s

7. Power factor of an L-R series circuit is 0.6 and that of a C-R series circuit is 0.5. If the elements (L, C and R) of the two circuit are joined in series the power factor of this circuit is found to be 1. The ratio of the resistance is L-R circuit to the resistance in the C-R circuit is

(A) $\frac{6}{5}$ (B) $\frac{5}{6}$
(C) $\frac{4}{3\sqrt{3}}$ (D) $\frac{3\sqrt{3}}{4}$

8. One mole of an ideal gas is kept enclosed under a light and frictionless piston (area = 10^{-4} m^2) connected by a compressed spring (spring constant 100 N/m). The volume of gas is 0.83 m^2 and its temperature is 100 K. The gas is heated so that it compresses the spring further by 0.1 m. The work done by the gas in the process is X joules. Find the value of X. (Take R = 8.3 J-mol^{-1}K^{-1} and there is no atmosphere)



(A) $\frac{1}{2}$ (B) $\frac{3}{2}$
(C) $\frac{5}{2}$ (D) $\frac{3}{4}$

9. A charged particle of specific charge α is released from origin at t = 0 with velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$ in magnetic field $\vec{B}_0 = B_0\hat{i}$. The coordinate of the particle at time $t = \frac{\pi}{B_0\alpha}$ are (specific charge α = q/m)

(A) $\left(\frac{v_0}{2B_0\alpha}, \frac{\sqrt{2}v_0}{\alpha B_0}, \frac{-v_0}{B_0\alpha}\right)$ (B) $\left(\frac{-v_0}{2B_0\alpha}, 0, 0\right)$

(C) $\left(0, \frac{2v_0}{\alpha B_0}, \frac{v_0\pi}{2B_0\alpha}\right)$ (D) $\left(\frac{v_0\pi}{B_0\alpha}, 0, \frac{-2v_0}{B_0\alpha}\right)$

10. An electron is released from the origin at a place where a uniform electric field $\vec{E}$ and uniform magnetic field $\vec{B}$ exists along the negative y-axis and negative z-axis respectively. Then which of the following are INCORRECT.

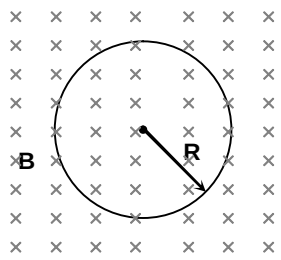
(A) At time t the y-component of velocity of the electron becomes $v_y = \frac{E}{B}\sin\omega t$, where $\omega = \frac{eB}{m}$

(B) At $t = \frac{\pi m}{eB}$ the electron will have only x-component of velocity.

(C) At $t = \frac{2\pi m}{3eB}$, the y-component of velocity becomes zero.

(D) The displacement along y-axis is $\frac{2Em}{eB^2}$, when the velocity of electron becomes perpendicular to the y-axis.

11. A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of the ring. If the radius R varies as a function of time 't', as $R = R_0 + t$, the emf induced in the loop is



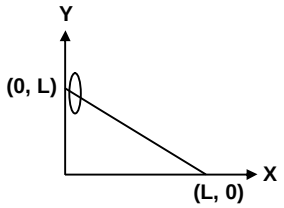
(A) $2\pi(R_0 + t)B$ clockwise

(B) $\pi(R_0 + t)B$ clockwise

(C) $2\pi(R_0 + t)B$ anticlockwise

(D) zero

12. Electric field given by the vector $\vec{E} = x\hat{i} + y\hat{j}$ present in the XY plane. A small ring carrying charge +Q, which can freely slide on smooth non-conducting rod, is projected along the rod from the point (0, L) such that it can reach the other end of the rod. What minimum velocity should be given to the ring? (Assume z-axis is the vertical axis)



(A) $\left(\frac{QL^2}{m}\right)^{1/2}$ (B) $2\left(\frac{QL^2}{m}\right)^{1/2}$

(C) $4\left(\frac{QL^2}{m}\right)^{1/2}$ (D) $\left(\frac{QL^2}{2m}\right)^{1/2}$

13. Current density in a cylindrical wire of radius R is given as

$$J=\begin{cases} J_0\left(\frac{x}{R}-1\right) & \text{for} \quad 0 \le x < \frac{R}{2} \\ J_0\left(\frac{x}{R}\right) & \text{for} \quad \frac{R}{2} \le x \le R \end{cases}$$

The current flowing in the wire is

(A) $\frac{7}{24}\pi J_0 R^2$ (B) $\frac{1}{6}\pi J_0 R^2$

(C) $\frac{7}{12}\pi J_0 R^2$ (D) $\frac{5}{12}\pi J_0 R^2$

14. A container encloses 32 g of O_2 at 127°C and of pressure P_0. A small hole is made in the container and the gas leaks out. At a certain state, pressure of gas becomes $\frac{P_0}{2}$ and its temperature 27°C. Mass of the gas that has leaked out till this stage is nearly:

(A) 6.3g (B) 7.4g

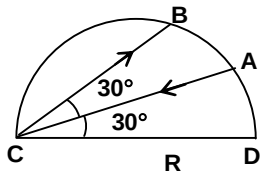
(C) 16.3g (D) 10.7g

15. A long straight wire along the z-axis carries I in the negative z-direction. The magnetic vector field B at a point having coordinates (x, y) in the z = 0 plane is

(A) $\frac{\mu_0 I(y\hat{i}-x\hat{j})}{2\pi(x^2+y^2)}$ (B) $\frac{\mu_0 I(y\hat{i}+x\hat{j})}{2\pi(x^2+y^2)}$

(C) $\frac{\mu_0 I(x\hat{i}-y\hat{j})}{2\pi(x^2+y^2)}$ (D) $\frac{\mu_0 I(x\hat{i}+y\hat{j})}{2\pi(x^2+y^2)}$

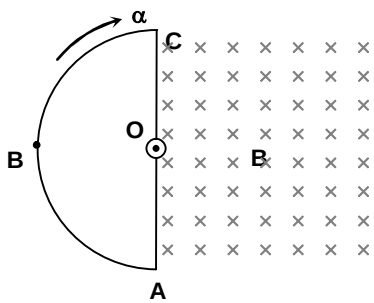
16. A current carrying wire is placed in the grooves of an insulating semi-circular disc of radius R, a shown. The current enters at point A and leaves from point B. The magnetic field at point D, has magnitude:



(A) $\frac{\mu_0 I}{8\pi R\sqrt{3}}$ (B) $\frac{\mu_0 I}{4\pi R\sqrt{3}}$

(C) $\frac{\sqrt{3}\mu_0 I}{4\pi R}$ (D) $\frac{\sqrt{3}\mu_0 I}{2\pi R}$

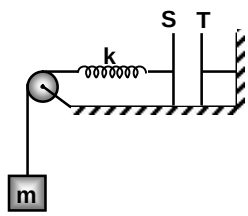
17. What is the emf induced in the loop OABCO (radius = a) at time t placed in uniform magnetic field B as shown when its angular acceleration is β? n = number of half revolutions of the loop in a given time t.



(A) $\frac{Ba^2\beta t}{2}$ (B) $\frac{(-1)^n Ba^2\beta t}{2}$

(C) $\frac{(-1)^n Ba^2\beta t}{4}$ (D) $(-1)^n Ba\beta t$

18. The plate S and T of an uncharged parallel plate capacitor are connected across a battery. The battery is then disconnected and the charged plates are now connected in a system as shown in the figure. The system shown is in equilibrium. All the strings are massless. The magnitude of charge on one of the capacitor plate is: (consider the area of plates to be A)



(A) $\sqrt{2mgA\varepsilon_0}$

(B) $\sqrt{\frac{4mgA\varepsilon_0}{k}}$

(C) $\sqrt{mgA\varepsilon_0}$

(D) $\sqrt{\frac{2mgA\varepsilon_0}{k}}$

19. Power radiated by a black body is P_0 and the wavelength corresponding to maximum energy is around λ_0. On changing the temperature of the black body, it was observed that the power radiated becomes $\frac{256P_0}{81}$. The shift in wavelength corresponding to the maximum energy will be

(A) $\frac{\lambda_0}{4}$

(B) $\frac{\lambda_0}{2}$

(C) $-\frac{\lambda_0}{4}$

(D) $-\frac{\lambda_0}{2}$

20. One mole of an ideal gas contained in a cylinder undergoes a thermodynamic process during which pressure relates to volume as $P = \frac{A}{1+\left(\frac{B}{V}\right)^2}$, where A and B are constants. As the volume of the gas is changed from V = B to V = 2B, its change of temperature can be expressed as:

(A) $\frac{6AB}{5R}$

(B) $\frac{AB}{2R}$

(C) $\frac{-3AB}{5R}$

(D) $\frac{11AB}{10R}$

SECTION – B

(Numerical Answer Type)

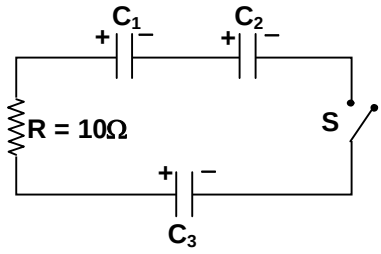
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A moving coil galvanometer 'A' has 100 turns and resistance 50 Ω. Another moving coil galvanometer 'B' has number of turns 500 and resistance 200 Ω. The other quantities are same in both cases. They are connected to a battery through a resistance of 50Ω. The ratio of deflection of B to A is …….

22. Three concentric thin spherical shells of radii R, 2R and 3R are given 2Q, Q and −Q respectively. The innermost and the outermost shells are earthed. The final charge on the inner most sphere is $\left(-\frac{Q}{n}\right)$. Find the value of n.

23. A galvanometer having a resistance of 50Ω, gives a full scale deflection for a current of 0.05 A. Find the length (in meter) of a resistance wire of area of cross-section 2.97×10^{-2} cm^2 that can be used to convert the galvanometer into an ammeter which can read a maximum of 5A current. (specific resistance of the wire = 5×10^{-7} Ω-m)

24. Three charged capacitors C_1, C_2 and C_3 of capacitances 3 μF, 6 μF and 6 μF respectively and having charge 30 μC each are connected with a resistor of resistance R = 10Ω through a switch as shown in the figure. The switch S is closed at t = 0. Find the total heat (in μJ) dissipated through the resistor.



25. Two identical containers A and B with frictionless pistons contain the same ideal gas at the same temperature and the same volume V. The mass of the gas in A is m_A and that in B is m_B. The gas in each cylinder is now allowed to expand isothermally to the same final volume 2V. The change in pressure in A and B are found to be ΔP and $1.5\Delta P$ respectively. If $\frac{m_A}{m_B} = \frac{2}{\lambda}$ then find the value of λ.

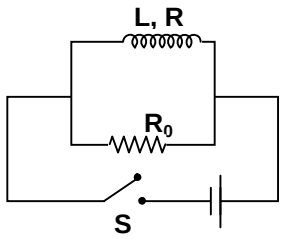
26. An ideal gas has a molar heat capacity $C = \frac{9R}{2}$ in the process PT = constant. Find the degree of freedom of the gas molecule. (R = ideal gas constant)

27. When an ideal diatomic gas is heated at constant pressure, during a process. If during the given process, the ratio $\frac{\Delta Q}{\Delta W} = \frac{n}{2}$. Find the value of n. (where ΔQ = Heat supplied to the gas during the process, ΔW = work done by the gas during the process)

28. If P-V diagram of a diatomic gas during a process is plotted, it is a straight line (when extended it passes through the origin). If molar heat capacity of the gas in the process is nR. Find the value of n. (where R = ideal gas constant)

29. In a region of space, the magnetic field is changing at a constant rate $\frac{dB}{dt}$. A given mass 'm' of copper of resistivity ρ and density d drawn into a wire of radius r and formed into a circular loop of radius R_1 which is placed perpendicular to the given magnetic field region. If the current induced in the loop in the terms of given quantities is $\frac{nm}{4\pi\rho d}\frac{dB}{dt}$. Find n.

30. A coil of inductance L = 2μH and R = 1Ω is connected to a source of constant emf, ε = 3V. A resistance R_0 = 2Ω is connected in parallel with the coil. The heat generated in the coil after the switch S is disconnected after a long time is $P \times 10^{-6}$ J. The internal resistance of the source is negligible. Find the value of P.



Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

31. Molecular formula of P and Q is C_2H_4O and C_3H_6O. If P and Q both give (+ve) $NaHSO_3$ test and only P gives haloform test. Choose the INCORRECT option:

$P + Q \xrightarrow[2.\ \Delta]{1.\ NaOH} Product$

(A) Total 4 products will be obtained excluding stereoisomers.
(B) Total 4 products will be obtained including stereoisomers.
(C) Reaction is known as cross-Aldol condensation.
(D) All products obtained will give (+ve) Tollen't test as well as Fehling test.

32. If Q is benzal chloride, then choose the INCORRECT statement about this commercial method

$P \xrightarrow[h\nu]{Cl_2} Q \xrightarrow[373\ K]{H_2O} R$

(A) R will give (–ve) Fehling test.
(B) Total 10 benzenoid structural isomers for Q are possible.
(C) (P) can have molecular formula C_7H_8.
(D) Free radical addition involves in conversion of P to Q.

33. The reaction in which two moles of (same or different) alcohol is not obtained as a product is

(A) $H_3C-\underset{\underset{O}{\|}}{C}-OC_2H_5 \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4}$

(B) $H_3C-\underset{\underset{O}{\|}}{C}-O-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4}$

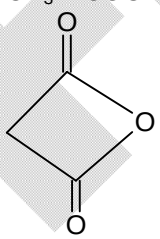
(C) $H_3C-\underset{\underset{O}{\|}}{C}-OCH_2CH_3 \xrightarrow[(ii)\ H_2O]{(i)\ DIBAL-H}$

(D) $H_3C-\underset{\underset{O}{\|}}{C}-OCH_3 \xrightarrow[C_2H_5OH]{Na}$

34. The product obtained on reaction of Malonic acid with $P_2O_5\ /\ \Delta$ is

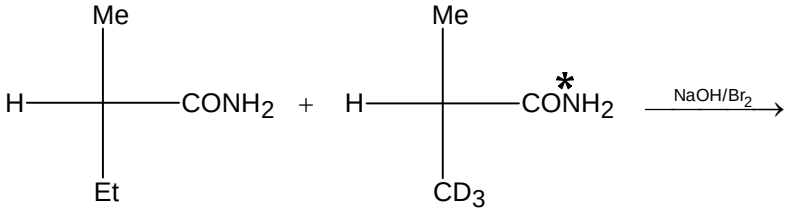
(A) $CH_3 - COOH$

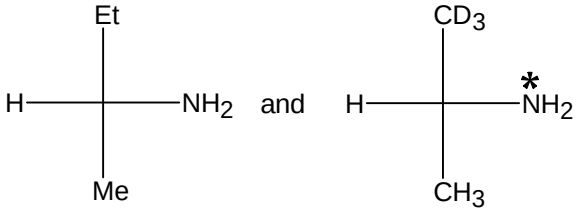
(B)

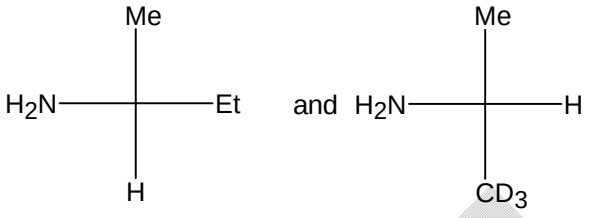


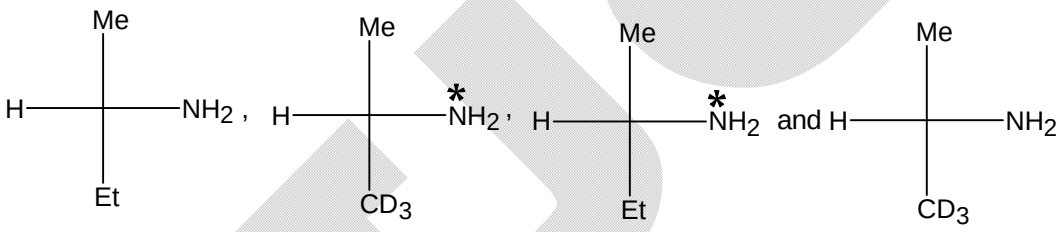
(C) O = C = C = C = O
(D) O = C = C = O

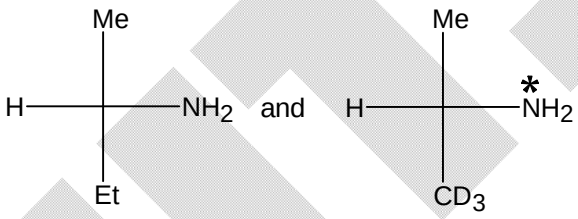
35. The product of the following reaction is:



(A) 

(B) 

(C) 

(D) 

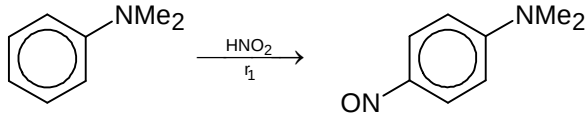
36. The correct order of rate of electrophilic aromatic substitution is

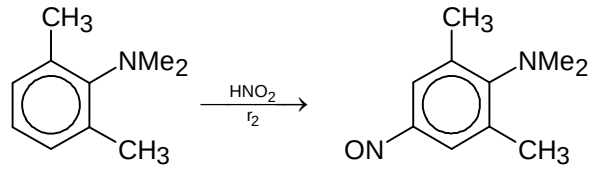


(I) (II) (III)

(A) I > II > III
(B) I < II < III
(C) II > I > III
(D) III > I > II

37. Consider the following reaction

Reaction - I 

Reaction - II 

Which of the following is a CORRECT comparison of rate of reaction?

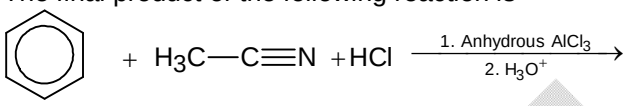
(A) $r_1 > r_2$

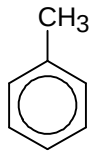
(B) $r_2 > r_1$

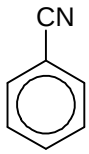
(C) $r_1 = r_2$

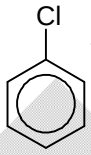
(D) Reactions are not possible

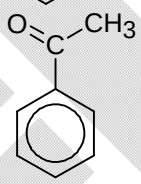
38. The final product of the following reaction is



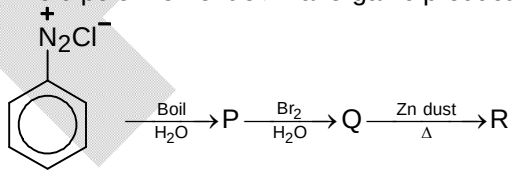
(A) 

(B) 

(C) 

(D) 

39. The dipole moment of final organic product obtained in the following reaction is:



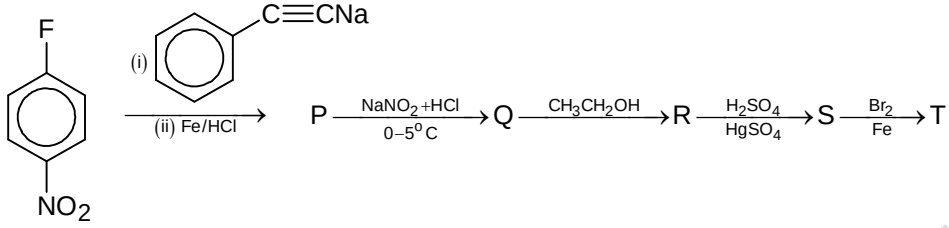
(A) $\mu_R \neq 0$

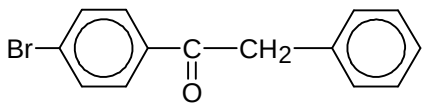
(B) $\mu_Q = 0$

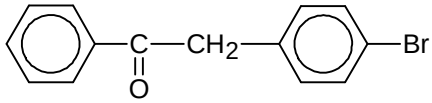
(C) $\mu_R < \mu_Q$

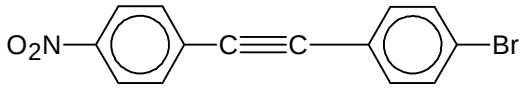
(D) $\mu_R > \mu_Q$

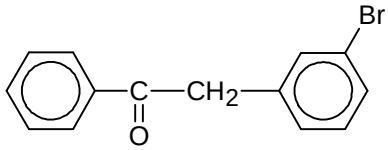
40. The final product 'T' in the following reaction sequence is



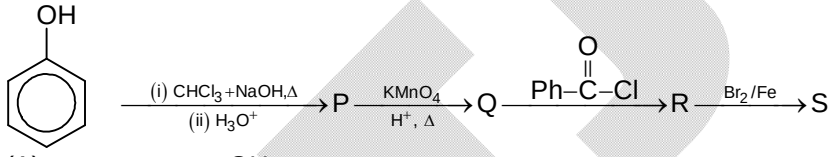
(A)


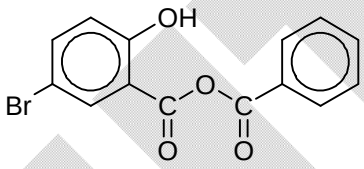
(B)


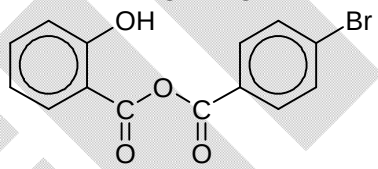
(C)


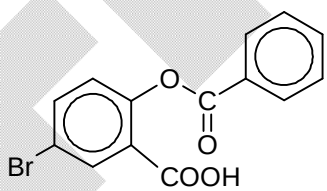
(D)


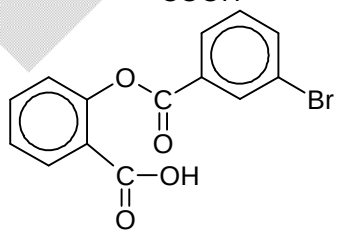
41. The major product 'S' obtained in the following reaction is



(A)


(B)


(C)


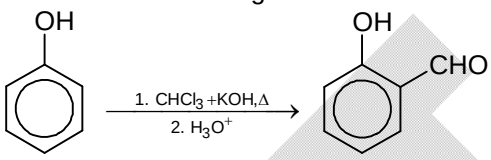
(D)


42. Compare van't Hoff factor for compounds CH_3CH_2OH (A) and Ph_3COH (B) in cold conc. H_2SO_4 determined from measurement of freezing point.
(A) $i_A > i_B$
(B) $i_B > i_A$
(C) $i_A = i_B$
(D) $i_A < i_B < 1$

43. 896 mL vapour of a hydrocarbon 'P' having carbon 87.80% and hydrogen 12.19% weighs 3.28 g at STP. Hydrogenation of 'P' gives 2-methyl pentane. Also 'P' on hydration in the presence of H_2SO_4 / $HgSO_4$ gives a ketone 'Q' having molecular formula $C_6H_{12}O$. The ketone 'Q' gives a positive iodoform test. The INCORRECT statement is:
(A) P gives white ppt. with Ammonical $AgNO_3$.
(B) Q gives orange colour with 2,4-DNP.
(C) Q gives crystalline compound with $NaHSO_3$.
(D) P gives red colour of Cu_2O with Ammonical Cu_2Cl_2.

44. Phenol and formaldehyde undergo condensation to give a polymer (P) which on heating with formaldehyde gives a thermosetting polymer (Q). The polymer (P) is:
(A) Novolac
(B) Bakelite
(C) Glyptal
(D) Polyster

45. Consider the following reaction



(Major)

The intermediate formed in above reaction is
(A) Singlet dichloro carbene
(B) $\overset{-}{C}Cl_3$
(C) Chloronium ion
(D) $\overset{\oplus}{C}Cl_2$

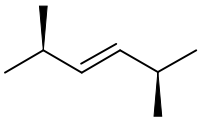
46. The basic strength order of the following in H_2O is

$MeNH_2$, Me_2NH , Me_3N
(I) (II) (III)

(A) I > II > III
(B) II > I > III
(C) III > II > I
(D) I > III > II

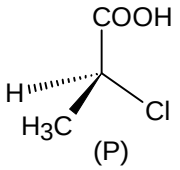
47. Most acidic compound among the following is
(A) Picric acid
(B) Squaric acid
(C) Carbonic acid
(D) Carbolic acid

48. The total number of stereoisomers of following compound 'X' is

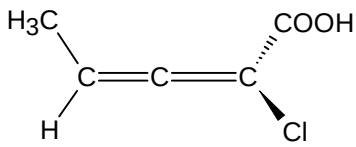


(A) 2
(B) 6
(C) 4
(D) 8

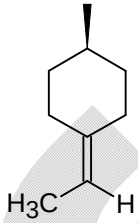
49. Asymmetric compound among the following is



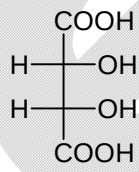
(P)



(Q)

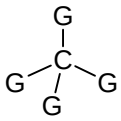


(R)



(S)

(A) P, Q, R, S
(B) P, Q, S
(C) P, Q, R
(D) Q, S, R

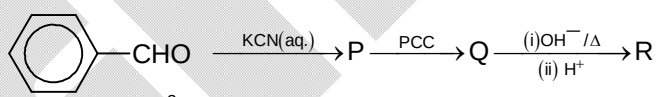
50. A compound  contain $\frac{\sigma}{\pi}$ ratio 1 : 1, if G is

(A) CHO
(B) CN
(C) COOH
(D) C ≡ CH

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. 

How many sp^2 hybridised carbons are present in [R]?

52. In soda lime decarboxylation, soda lime is made of x mole of NaOH and y moles of CaO, then $\frac{x}{y}$ is….

53. 0.5 g of an organic compound having molecular formula $C_{12}H_{24}N_x$ was analysed for nitrogen using Duma's method. The nitrogen evolved was collected over water and found to be 110 mL at 300 K and 802 mm Hg pressure. If aqueous tension of H_2O at 300 K is 42 mm Hg, the value of x is…..

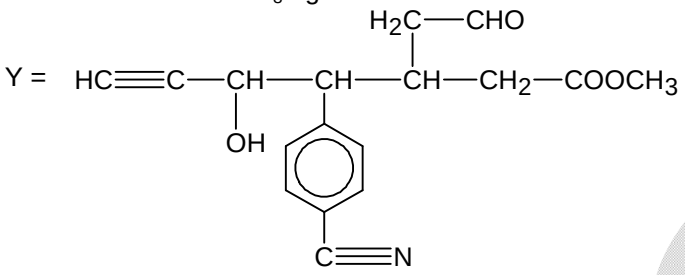
54. In the qualitative analysis of sulphide ions $\left(S^{-2}\right)$, the following reaction is used

$$S^{-2} + \text{Compound}(X) \longrightarrow Na_4\left[Fe(CN)_5(NOS)\right]$$

What is magnetic moment (in B. M.) for compound 'X'?

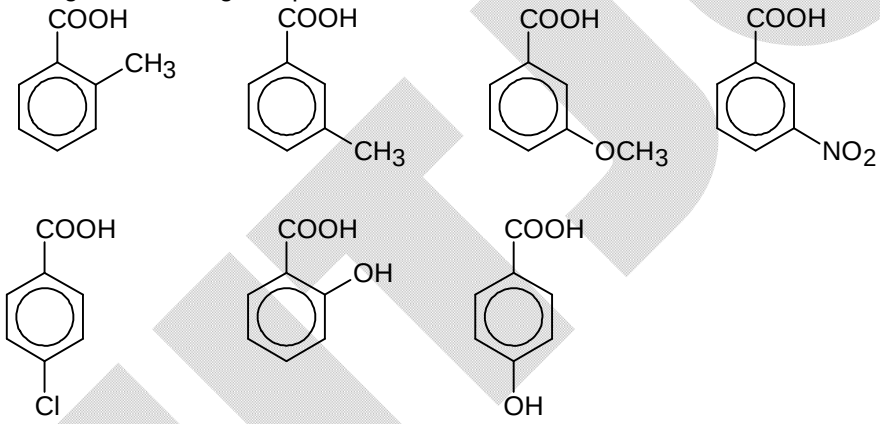
55. How many alcoholic structural isomers of molecular formula $C_6H_{14}O$ are possible which on oxidation with PCC give aldehyde product?

56. Number of moles of CH_3MgBr which will react with one mole of compound 'Y':

Y = 

57. An organic compound 'A' having molecular formula $C_5H_{10}O_6$ on acetylation forms a compound having molecular weight 334. Number of hydroxyl groups (–OH) in the compound 'A' is:

58. Among the following compounds, which are more acidic than benzoic acid



59. Total number of alkenes of molecular formula C_6H_{12} which will show geometrical isomerism

60. Total number of meso structures in $CH_3(CHOH)_5CH_3$ is

Mathematics

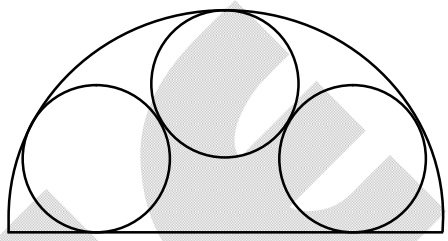
PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

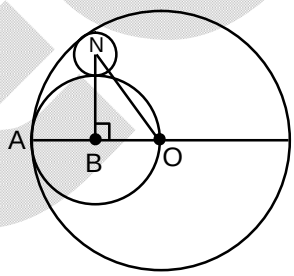
61. Three identical circles, each of radius a, are drawn as shown in figure and tangent to a semi-circle of radius r, express a in terms of r ?



(A) $\frac{r}{4}$ (B) $\frac{r}{2}$

(C) $\frac{r}{3}$ (D) $\frac{2r}{3}$

62. In the figure below is a circle with centre O, and another circle with centre at the midpoint B of the radius OA and passing through O. A small circle is drawn as shown, tangent to both the larger circles, such that its centre N is directly above B (that is, BN $\perp$ AB). If the perimeter of the triangle NBA is 8 cm, find the value (in cm) of the radius of the small circle



(A) $\frac{2}{3}$ (B) 2

(C) 1 (D) $\sqrt{2}$

63. In a ΔABC
$3\sin A + 4\cos B = 6$
and $4\sin B + 3\cos A = 1$
then angle C is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$ (D) $\frac{2\pi}{3}$

64. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$, then $f_4(x) - f_6(x) = \frac{\lambda}{24}$, then λ is

(A) 1 (B) 2

(C) -2 (D) -1

65. In a triangle ABC, $\frac{a}{b} = 2 + \sqrt{3}$ and $\angle C = 60°$, then

(A) $\angle A = 15°$ (B) $\angle A = 105°$

(C) $\angle B = 105°$ (D) $\angle B = 30°$

66. Let $A(\alpha, 2)$, $B(\alpha, -2)$ and $C(\beta, 2)$ be the vertices of a triangle ABC. If the circumcentre is origin and orthocentre is $(3, \gamma)$, then which of the following is **NOT CORRECT** about ΔABC

(A) area is 12 (B) perimeter is $10 + 2\sqrt{13}$

(C) inradius is $\frac{5 - \sqrt{13}}{2}$ (D) circum radius is $2\sqrt{13}$

67. Let PQ be a chord of length 8 of the circle $4x^2 + 4y^2 + 24x - 32y - 189 = 0$. If tangents drawn to the circle at point P and Q intersect at the point R, then perpendicular distance of R from chord PQ is

(A) $\frac{15}{2}$ (B) $\frac{32}{15}$

(C) $\frac{68}{15}$ (D) $\frac{36}{17}$

68. Let P_1 be a parabola with vertex (1, 8) and focus (3, –1) and P_2 is the mirror image of P_1 with respect to the line $3x - y + 10 = 0$. If the focus of P_2 is (α, β) and directrix of P_2 is $7x + 6y = \gamma$, then $\alpha + \beta + \gamma$ is equal to

(A) 125 (B) 137

(C) 52 (D) 119

69. If two tangents can be drawn to the different branches of hyperbola $x^2 - 9y^2 = 9$ from the point (α, β) then (α, β) can be

(A) $\left(2, \frac{4}{9}\right)$ (B) $\left(-1, \frac{1}{4}\right)$

(C) $\left(1, \frac{1}{2}\right)$ (D) (5, 1)

70. Let the line $y = mx$ and the ellipse $4x^2 + 9y^2 = 1$ intersect at point P in the first quadrant. If the normal to the ellipse at P meets the coordinate axes at A(α, 0) and B$\left(0, -\frac{5}{8\sqrt{3}}\right)$, then area of triangle OAB is equal to (where O is origin)

(A) $\frac{1}{16\sqrt{3}}$ (B) $\frac{25}{576\sqrt{3}}$

(C) $\frac{9}{144\sqrt{3}}$ (D) $\frac{25}{288\sqrt{3}}$

71. Consider the circle $x^2 + y^2 = 25$ and $y^2 = 24x$. They intersect at P and Q in the first and fourth quadrant. Tangents at P and Q to the circle intersect the x-axis at R and tangent at P and Q to the parabola intersect x-axis at S, then radius of circumcircle of ΔPRS is

(A) $20\sqrt{7}$ (B) $5\sqrt{7}$

(C) $2\sqrt{7}$ (D) $7\sqrt{3}$

72. Perpendicular tangents are drawn from the point P to an ellipse (whose axes are parallel to coordinate axes). The locus of point P is $x^2 + y^2 - 4x - 6y - 5 = 0$ and if the product of length of perpendicular from foci upon any one of these tangents is 6, then eccentricity is

(A) $\frac{1}{\sqrt{6}}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$

73. If the directrix of hyperbola (having positive x intercept) $xy = 8 + x$ is the tangent to the hyperbola $x^2 - ky^2 = 30$, then k is equal to

(A) $\sqrt{5}$ (B) 6

(C) $\sqrt{6}$ (D) $\sqrt{15}$

74. Given points P, Q on an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b > 0) satisfying OP $\perp$ OQ, the minimum of |OP| × |OQ| is

(A) $\frac{2a^2b^2}{a^2+b^2}$ (B) $\frac{2a^2b^2}{a^2-b^2}$

(C) $\frac{4a^2b^2}{a^2-b^2}$ (D) $\frac{4a^2b^2}{a^2+b^2}$

75. The straight line $\frac{x}{4}+\frac{y}{3}=1$ intersect the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at two points A and B, there is a point P on the ellipse such that the area of ΔPAB is equal to 3. Then the number of possible point P is/are

(A) 1 (B) 2

(C) 4 (D) 3

76. Suppose points F_1, F_2 are the focii of ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, P is a point on the ellipse such that $\frac{|PF_1|}{|PF_2|}=\frac{2}{1}$, then the area of ΔPF_1F_2 is equal to

(A) 16 (B) 20

(C) 4 (D) 32

77. The curve represented by the equation $\frac{x^2}{\sin\sqrt{2}-\sin\sqrt{3}}+\frac{y^2}{\cos\sqrt{2}-\cos\sqrt{3}}=1$ is

(A) an ellipse with the focii on x-axis (B) an ellipse with the focii on y-axis

(C) an hyperbola with focii on x-axis (D) an hyperbola with focii on y-axis

78. If one side of square ABCD is on the line y = 2x – 17 and the other two vertices lies on the parabola $y = x^2$, then the minimum area of the square is

(A) 160 (B) 1200

(C) 80 (D) 60

79. Let a line with the inclination of 60° from positive x-axis be drawn through the focus F of the parabola $y^2 = 8(x + 2)$. If the two intersection points of the line and the parabola are A and B and the perpendicular bisector of the chord AB intersect the x-axis at the point P, then the length of segment PF is

(A) $\frac{16}{3}$ (B) $\frac{8}{3}$

(C) $\frac{16\sqrt{3}}{3}$ (D) $8\sqrt{3}$

80. ΔABC is inscribed in a unit circle. The three bisectors of the angles A, B, C are extended to intersect the circle at A_1, B_1, C_1 respectively, then the value of $\frac{AA_1\cos\frac{A}{2}+BB_1\cos\frac{B}{2}+CC_1\cos\frac{C}{2}}{\sin A+\sin B+\sin C}$ is

(A) 2 (B) 4

(C) 6 (D) 8

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $x_0 = 2022$ and let $x_{n+1} = \frac{1+x_n}{1-x_n}$ for $n \geq 1$, find x_{2024}

82. Let $x + y + z = 1$, $x, y, z \in R^+$ then determine the least value of $\frac{1}{x}+\frac{4}{y}+\frac{9}{z}$.

83. The value of $(1 - \cot 23°)(1 - \cot 22°)$ is equal to

84. The number of values of x in the interval $\left(0,\frac{\pi}{2}\right)$ such that $\frac{\sqrt{3}-1}{\sin x}+\frac{\sqrt{3}+1}{\cos x}=4\sqrt{2}$ is

85. If $\alpha = \frac{\pi}{2023}$, then value of $\cos\alpha.\cos2\alpha.\cos3\alpha.\cos4\alpha \ldots\ldots \cos(1011\alpha) = 2^{-\lambda}$, then λ is

86. A horizontal park is in the shape of a rectangle ABCD with AB = 30m. A vertical lamp post BP is created at the point B such that $\angle PCB = 30°$ and $\angle PDB = \tan^{-1}\left(\frac{8}{17\sqrt{3}}\right)$. If height of the lamp post is h, then [h] is equal to (where [.] denotes G.I.F.)

87. If the asymptotes of the hyperbola $9x^2 - 4y^2 - 18x - 24y - 63 = 0$ are tangents to the circle $13x^2 + 13y^2 - 26x + k = 0$ at point $A(\alpha_1, \beta_1)$ and $B(\alpha_2, \beta_2)$ then $|\alpha_1 + \alpha_2 + k|$ is equal to

88. Let α and β be real numbers in the interval $\left[0, \frac{\pi}{2}\right]$, then of $\sin^6\alpha + 3\sin^2\alpha\cos^2\beta + \cos^6\beta = 1$, then $\frac{\alpha}{\beta}$ is equal to

89. The points (0, 0), (a, 11) and (b, 37) are the vertices of an equilateral triangle, then ab is equal to

90. Let P is a point, lies on circle $(x + 5)^2 + (y - 12)^2 = 196$, then the minimum value of sum of squares of abscissa and ordinate of point P is